Information Security · Lecture 4 — Substitution Techniques

Complete prep + practice · Monoalphabetic · Caesar · Playfair · Polyalphabetic · Vigenère · Vernam · Hill · follows the lecture's conventions (A = 0) · every answer verified by computation

Part 0 — Toolkit (memorise first)

Alphabet table (A = 0)

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
13	14	15	16	17	18	19	20	21	22	23	24	25

💡 Anchors: **A=0, E=4, J=9, O=14, T=19, Z=25** — *"Every Jolly Owl Talks Zen"*.

Three golden rules of mod 26

1. **Bigger than 25?** Subtract 26 (or a multiple) → 29 → 3.
2. **Negative? Add 26** → −9 + 26 = 17.
3. The answer is always **0 – 25**.

Multiples of 26 (for Hill): 26 · 52 · 78 · 104 · 130 · 156 · 182 · 208 · 234 · 260 · 286 · 312 · 338 · 364 · 390 · 416 · 442 · 468 · 494 · 520 (e.g. 114 mod 26 = 114 − 104 = 10)

Part 1 — The Big Picture

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Symmetric (conventional) encryption
├── Substitution  → letters are REPLACED
│   ├── Monoalphabetic (one fixed alphabet, one-to-one)
│   │    ├── Caesar cipher
│   │    └── Playfair cipher      ← teacher's classification
│   └── Polyalphabetic (many alphabets, one-to-many)
│        ├── Vigenère cipher
│        ├── Vernam cipher (One-Time Pad)
│        └── Hill cipher          ← teacher's classification
└── Transposition → letters are REARRANGED
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⚠ Some textbooks call Playfair a "digraph" and Hill a "polygraphic" cipher. **In the paper, follow the slides:** Playfair = monoalphabetic, Hill = polyalphabetic.

	Monoalphabetic	Polyalphabetic
Alphabets	One fixed	Multiple
Mapping changes?	Never	Changes during message
Relationship	One-to-one	One-to-many
Same plain letter →	Always same cipher letter	Can become different letters
Slide example	MY NAME → **N**P OB**N**Z	MY NAME → **N**P OB**X**Z
Frequency analysis	Vulnerable	Reduces patterns
Strength	Weaker	Stronger
Examples	Caesar, Playfair	Vigenère, Vernam, Hill

1. Monoalphabetic Substitution Cipher (Slide 18)

Uses **one fixed substitution alphabet** for the whole message; each plaintext letter is replaced by **one specific** ciphertext letter and the substitution **never changes**.

- One-to-one relationship · M always becomes N (MY NAME → NP OBNZ).
- **Weakness:** frequency analysis — letter patterns remain (E is the most common English letter).
- **Examples:** Caesar, Playfair.

General monoalphabetic key (used in practice M1–M3):

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

Practice — Monoalphabetic

1. Encrypt **SECRET** with the key table above.
2. Encrypt **MONEY**.
3. Decrypt **ITSSG**.
4. In "MY NAME → NP OBNZ", which cipher letter appears twice and why does that prove it is monoalphabetic?
5. Why is a monoalphabetic cipher weak against frequency analysis?
6. True/False: "In a monoalphabetic cipher, A can be encrypted as D in one place and K in another."

2. Caesar Cipher (Slides 20–25)

- Also called the **Shift Cipher**; each letter shifted by a fixed number; **monoalphabetic**; one of the earliest ciphers.
- **Julius Caesar used key = 3** for military messages.

Encryption: C = E(k, p) = (p + k) mod 26 **Decryption:** P = D(k, c) = (c − k) mod 26 (negative → add 26)
💡 *"Plus to hide, minus to find."*

Key space: 1 ≤ k ≤ 25 → only **25 keys** (k = 0 gives the original text) → **brute-force attack** tries all 25.

Table method (slide 24): HELLO, k = 4

Letter	Value	C = (p + k) mod 26	Result	Cipher
H	7	(7+4) mod 26 = 11	11	L
E	4	(4+4) mod 26 = 8	8	I
L	11	(11+4) mod 26 = 15	15	P
L	11	(11+4) mod 26 = 15	15	P
O	14	(14+4) mod 26 = 18	18	S

Ciphertext: **LIPPS**. Decryption uses the same table with (c − k).

⚠ Traps: X + 3 = 26 → 26 mod 26 = 0 = **A** · D − 4 = −1 → +26 = 25 = **Z** · Slide 21 "A = 1 implies B" means **key 1**, values still start at A = 0 · ignore spaces.

Practice — Caesar

1. Encrypt **ATTACK**, k = 3.
2. Encrypt **ZEBRA**, k = 5 (wrap-around).
3. Decrypt **WKLV LV IXQ**, k = 3.
4. Decrypt **BQQMF**, k = 1.
5. Decrypt **EYHF**, k = 7 (negative values).

6. Ciphertext **KHOOR**, key unknown — brute force to find key and plaintext.
7. Encrypt **SECURITY**, k = 10, using the full table method.
8. Why are there only 25 useful keys? What attack does this allow?

3. Playfair Cipher (Slides 26–29)

- Invented in **1854 by Wheatstone** · **5 × 5 matrix** · **I and J share one cell** · encrypts **pairs (digraphs)**.

Step 1 — Build the matrix

Write the keyword letters first (skip repeats), then the remaining alphabet in order (I/J together). Must have exactly 25 cells. Slide key **ABHI**:

A	**B**	**H**	**I/J**	**C**
D	**E**	**F**	**G**	**K**
L	**M**	**N**	**O**	**P**
Q	**R**	**S**	**T**	**U**
V	**W**	**X**	**Y**	**Z**

Step 2 — Make pairs (teacher's rules)

Situation	Rule	Example
Normal	Split into pairs	heyar → he ya r...
Lone last letter	Add **Z**	heyar → he ya **rz**
Same letter in a pair	Insert **X** after the first	hello → he **lx** lo

⚠ Teacher uses **Z** for a lone letter and **X** for doubles. After inserting X, **re-pair the rest**: BALLOON → BA LX LO ON. BOOKKEEPER → BO OK KE EP ER (no X needed — only pairs matter).

Step 3 — Encrypt each pair

Rule	Situation	Encrypt	Decrypt
1	Same row	Letter to the **right** (wrap)	Letter to the **left**
2	Same column	Letter **below** (wrap)	Letter **above**
3	Rectangle	Each letter → **other corner in its own row** (same both ways)	

💡 Rectangle: *"Stay in your row, move to your partner's column."*

Slide 28 worked (key ABHI, KSKLBMRWUQQW)

Pair	Rule	Result
KS	Rectangle: K → F, S → U	**FU**
KL	Rectangle: K → D, L → P	**DP**
BM	Same column, below: B → E, M → R	**ER**
RW	Same column: R → W, W wraps to B	**WB**
UQ	Same row, right: U wraps to Q, Q → R	**QR**
QW	Rectangle: Q → R, W → V	**RV**

Ciphertext **FUDPERWBQRRV** ✓

Slide 29 (second key matrix): hello → he lx lo → EC QZ BX → **ECQZBX** ✓

Checklist: skip repeated key letters · combine I/J · 25 cells · X for doubles and re-pair · Z for lone letter · rectangle = own row · wrap at edges.

Practice — Playfair

1. Key **MONARCHY**: build the matrix and encrypt **INSTRUMENTS**.
2. Key **ABHI**: encrypt **MEET**.
3. Key **SECURITY**: encrypt **BALLOON**.
4. Key **KEYWORD**: encrypt **HAMMER**.
5. Key **ABHI**: decrypt **ICEF**.
6. Digraphs only: (a) BOOKKEEPER (b) TOOTH (c) COMMITTEE.
7. Key **PLAYFAIR**: encrypt **HIDE THE GOLD**.
8. Why are I and J combined? Who invented Playfair, and when?

4. Polyalphabetic Cipher (Slide 19)

- Uses **multiple substitution alphabets**; encryption **changes during the message**; **no fixed substitution**.
- Same plain letter → different cipher letters (M → N, later M → X) · **one-to-many**.
- **Stronger** than monoalphabetic — **reduces frequency patterns**. Examples: Vigenère, Vernam, Hill.

Practice — Polyalphabetic

1. Define a polyalphabetic cipher.
2. In "MY NAME → NP OBXZ", how does M show it is polyalphabetic?
3. Why is polyalphabetic stronger than monoalphabetic?
4. Classify: Caesar, Vigenère, Playfair, Hill, Vernam.
5. One-to-one vs one-to-many: which is which?

5. Vigenère Cipher (Slides 30–35)

- Designed by **Blaise de Vigenère** (16th-century French) · **polyalphabetic** · 26 × 26 table or maths.
- **Repeat the key** until it is as long as the plaintext (remove spaces first).

Method 1 (table): encryption — plaintext letter in the first row, key letter in the first column, ciphertext where they meet (G × L → R). Decryption — go to the key's row, find the cipher letter, read its column header.
Method 2 (maths) ⭐: C = (P + K) mod 26 · P = (C − K) mod 26 (negative → +26)

Slide 31: GIVE MONEY, key LOCK

P	G	I	V	E	M	O	N	E	Y
P value	6	8	21	4	12	14	13	4	24
K	L	O	C	K	L	O	C	K	L
K value	11	14	2	10	11	14	2	10	11
Sum	17	22	23	14	23	28	15	14	35
mod 26	17	22	23	14	23	2	15	14	9
C	**R**	**W**	**X**	**O**	**X**	**C**	**P**	**O**	**J**

Ciphertext **RWXOXCPOJ** ✓ · Slide 35: "She is listening", key PASCAL → **HHWKSWXSLGNTCG** ✓

⚠ Remove spaces before repeating the key · write the key directly under the plaintext · 28 → 2 · negative → +26 · key length 1 = Caesar.

Practice — Vigenère

1. Encrypt **ATTACK AT DAWN**, key **LEMON**.
2. Encrypt **HELLO**, key **KEY**.
3. Decrypt **ARHIZVYM**, key **INFO**.
4. Encrypt **INFORMATION SECURITY**, key **PASCAL**.
5. Decrypt **MVWZXMQVYZLW**, key **KEY**.
6. If the key is one letter, which cipher does Vigenère become? Why is Vigenère stronger than Caesar?

6. Vernam Cipher / One-Time Pad (Slides 36–38)

- Letters → numbers (a = 0 ... z = 25) · **key length = plaintext length** · random key used once → **One-Time Pad**, theoretically unbreakable · combines plaintext and key using **XOR**.

⚠ Slide 36 says XOR, but the worked example (slides 37–38) uses **(P + K) mod 26**. Letters given → use mod 26. Binary bits given → use XOR.

Slide example: RANSWARDPK, key RANCHOBABA

P.T	R	A	N	S	W	A	R	D	P	K
value	17	0	13	18	22	0	17	3	15	10
Key	17	0	13	2	7	14	1	0	1	0
Sum	34	0	26	20	29	14	18	3	16	10
mod 26	8	0	0	20	3	14	18	3	16	10
C.T	**I**	**A**	**A**	**U**	**D**	**O**	**S**	**D**	**Q**	**K**

Ciphertext **IAAUDOSDQK** ✓ · Decryption: C − K, e.g. 8 − 17 = −9 → +26 = 17 = R ✓

XOR rules: $0 \oplus 0 = 0$, $1 \oplus 1 = 0$, $0 \oplus 1 = 1$, $1 \oplus 0 = 1$ — *"same = 0, different = 1"*. Decrypt with the same XOR: $(P \oplus K) \oplus K = P$.

	Vigenère	**Vernam**
Key length	Short, repeated	= message length
Repeats?	Yes	No
Security	Breakable (pattern)	Theoretically unbreakable

Practice — Vernam

1. Encrypt **HELLO**, key **XMCKL**.
2. Decrypt **EQNVZ**, key **XMCKL**.
3. Encrypt **MONEY**, key **BURNS**.
4. Decrypt **ROTQ**, key **ZOOM** (negatives).
5. Encrypt **ATTACK**, key **QWERTY**.
6. Binary: P = 1011 0110, K = 0110 1100. Find C with XOR, then decrypt C.
7. Why must the key be as long as the message? Why "one-time"?

7. Hill Cipher (Slides 39–46)

- Developed by **Lester Hill (1929)** · **polyalphabetic** · encrypts **blocks** simultaneously (Playfair only does pairs) · based on **linear algebra / matrices**.

Encryption: $C = K \cdot P \bmod 26$ **Decryption:** $P = K^{-1} \cdot C \bmod 26$
2×2 key → split plaintext into pairs → column vectors → multiply → mod 26.
$\begin{vmatrix} a & b \end{vmatrix} \times \begin{vmatrix} x \end{vmatrix} = \begin{vmatrix} a{\cdot}x + b{\cdot}y \end{vmatrix}$ $\begin{vmatrix} c & d \end{vmatrix} \times \begin{vmatrix} y \end{vmatrix} = \begin{vmatrix} c{\cdot}x + d{\cdot}y \end{vmatrix}$ — *"row times column"*

Slide example: ATTACK, K = |2 3; 3 6|

Pair	Vector	Top 2x + 3y	Bottom 3x + 6y	mod 26	Cipher
AT	(0, 19)	57	114	5, 10	**FK**
TA	(19, 0)	38	57	12, 5	**MF**
CK	(2, 10)	34	66	8, 14	**IO**

ATTACK → **FKMFIO** ✓

Finding K^{-1} — 5-step recipe

Step	What to do	Slide K = |2 3; 3 6|
1	Determinant d = ad − bc (mod 26)	12 − 9 = **3**
2	Find d^{-1}: d × d^{-1} mod 26 = 1	3 × 9 = 27 → **9**
3	Adjoint: swap a and d, negate b and c	|6 −3; −3 2|
4	Remove negatives (+26)	|6 23; 23 2|
5	K^{-1} = d^{-1} × adj, mod 26	9 × … = |54 207; 207 18| → **|2 25; 25 18|**

Inverses mod 26: 1↔1 · 3↔9 · 5↔21 · 7↔15 · 11↔19 · 17↔23 · 25↔25

Decrypt FKMFIO: FK (5,10) → 260, 305 → 0, 19 = AT · MF (12,5) → 149, 390 → 19, 0 = TA · IO (8,14) → 366, 452 → 2, 10 = CK → **ATTACK** ✓

⚠ Extra: if the determinant is even or 13 (shares a factor with 26) there is no inverse → the key is invalid (e.g. |2 1; 4 3|, det = 2).
Checklist: K × P (key first) · top row → top result · det = ad − bc · adjoint swaps a↔d, negates b and c · remove negatives before × d^{-1} · mod 26 at every step · pad an odd-length message.

Practice — Hill

1. Encrypt **HELP**, K = |3 3; 2 5|.
2. Find K^{-1} for K = |3 3; 2 5| (all 5 steps).
3. Decrypt **HIAT** using your K^{-1}.
4. Encrypt **CODE** with K = |2 3; 3 6|, then decrypt with K^{-1} = |2 25; 25 18| to check.
5. Encrypt **HELP** with K = |2 3; 3 6|.
6. K = |5 8; 17 3|: (a) det mod 26 (b) K^{-1} (c) encrypt **MATH**.
7. Is K = |2 1; 4 3| a valid key? Why?
8. Find the inverse mod 26 of 5, 7 and 11.

Part 2 — Master Comparison Table

Cipher	Type	Key	Encrypt	Decrypt	Special	Weakness / strength
Caesar	Mono	1–25	(p + k) mod 26	(c − k) mod 26	Caesar used 3	25 keys → brute force
Playfair	Mono (teacher)	Keyword → 5×5	Row right · column below · rectangle corners	Row left · column above · corners	I/J; X doubles, Z lone	Pairs; Wheatstone 1854
Vigenère	Poly	Keyword repeated	(P + K) mod 26	(C − K) mod 26	16th c. French	Repeating key pattern
Vernam	Poly	Length = message	(P + K) mod 26 / XOR	(C − K) mod 26 / XOR	One-Time Pad	Theoretically unbreakable
Hill	Poly	Matrix	C = KP mod 26	P = K^{-1}C mod 26	det needs inverse	Lester Hill 1929; blocks

Part 3 — Answer Key

Monoalphabetic

M1 **LTEKTZ** · M2 **DGFTN** · M3 **HELLO**
M4 **N** appears twice because M appears twice — same plain letter always gives the same cipher letter (one-to-one).
M5 Letter frequencies are preserved; the most common cipher letter is probably E, so the mapping can be recovered.
M6 **False**.

Caesar

C1 **DWWDFN** · C2 **EJGWF** (Z: 25 + 5 = 30 → 4 = E) · C3 **THIS IS FUN** · C4 **APPLE** · C5 **XRAY** (E: 4 − 7 = −3 → 23 = X)
C6 key = **3**, plaintext **HELLO**
C7 **COMEBSDI** (S 28→2 C · E→O · C→M · U 30→4 E · R→B · I→S · T 29→3 D · Y 34→8 I)
C8 k = 0 (or 26…) returns the original text, so only 1–25 are useful → **brute-force attack**.

Playfair

P1

M	**O**	**N**	**A**	**R**
C	**H**	**Y**	**B**	**D**
E	**F**	**G**	**I/J**	**K**
L	**P**	**Q**	**S**	**T**
U	**V**	**W**	**X**	**Z**

IN ST RU ME NT SZ → **GA TL MZ CL RQ TX** → **GATLMZCLRQTX**

P2 ME ET → **RM GR**

P3

S	**E**	**C**	**U**	**R**
I/J	**T**	**Y**	**A**	**B**
D	**F**	**G**	**H**	**K**
L	**M**	**N**	**O**	**P**
Q	**V**	**W**	**X**	**Z**

BA LX LO ON → **IB OQ MP PO** → **IBOQMPPO**

P4

K	**E**	**Y**	**W**	**O**
R	**D**	**A**	**B**	**C**
F	**G**	**H**	**I/J**	**L**
M	**N**	**P**	**Q**	**S**
T	**U**	**V**	**X**	**Z**

HA MX ME RZ → **PH QT NK CT** → **PHQTNKCT**

P5 IC EF → **HI DE** → **HIDE**
P6 (a) **BO OK KE EP ER** (b) **TO OT HZ** (c) **CO MX MI TX TE EZ**

P7

P	**L**	**A**	**Y**	**F**
I/J	**R**	**B**	**C**	**D**
E	**G**	**H**	**K**	**M**
N	**O**	**Q**	**S**	**T**
U	**V**	**W**	**X**	**Z**

HI DE TH EG OL DZ → **EB IM QM GH VR MF** → **EBIMQMGHVRMF**
P8 25 cells for 26 letters, so I and J share a cell. Invented by **Wheatstone in 1854**.

Polyalphabetic

PA1 Multiple substitution alphabets; mapping changes during the message; no fixed substitution.
PA2 M → N the first time and → X the second time.
PA3 It reduces frequency patterns, making frequency analysis harder.
PA4 Mono: Caesar, Playfair · Poly: Vigenère, Hill, Vernam.
PA5 One-to-one = monoalphabetic · one-to-many = polyalphabetic.

Vigenère

V1 **LXFOPVEFRNHR** · V2 **RIJVS** · V3 **SECURITY** · V4 **XNXQRXPTAQNDTCMTIEN** · V5 **CRYPTOGRAPHY**
V6 It becomes **Caesar**. Vigenère uses multiple alphabets, so the same letter encrypts differently depending on the key letter, hiding frequencies.

Vernam

VN1 **EQNVZ** · VN2 **HELLO** (E: 4 − 23 = −19 → 7 = H) · VN3 **NIERQ** · VN4 **SAFE** (R: 17 − 25 = −8 → 18 = S) · VN5 **QPXRVI**
VN6 C = **1101 1010**; C ⊕ K = 1011 0110 = P ✓
VN7 A shorter key would repeat and create patterns; "one-time" because each random key is used only once — then it is theoretically unbreakable.

Hill

> H1 HE (7,4): 33 → 7 (H), 34 → 8 (I) · LP (11,15): 78 → 0 (A), 97 → 19 (T) → **HIAT**
> H2 det = 15 − 6 = **9**; 9^{-1} = **3**; adj = |5 −3; −2 3| → |5 23; 24 3|; × 3 = |15 69; 72 9| → **K^{-1} = |15 17; 20 9|**
> H3 **HELP**
> H4 CODE → **UMSH**; check UM (20,12): 340 → 2 (C), 716 → 14 (O) ✓
> H5 **ATPT**
> H6 (a) det = 15 − 136 = −121 → **9** (b) 9^{-1} = 3 → **K^{-1} = |9 2; 1 15|** (c) MATH → **IWVG**
> H7 **No** — det = 6 − 4 = 2 (even) → no inverse mod 26 → cannot decrypt.
> H8 5^{-1} = **21**, 7^{-1} = **15**, 11^{-1} = **19**

Study Plan

1. **Day 1:** toolkit + Caesar + Monoalphabetic.
2. **Day 2:** Playfair — build every matrix by hand, P1–P8.
3. **Day 3:** Vigenère + Vernam (table layout every time).
4. **Day 4:** Hill — H1–H8 twice, memorise the inverse table.
5. **Day 5:** redo the slide examples without looking: HELLO → LIPPS, ABHI Playfair, GIVE MONEY, RANSWARDPK, ATTACK → FKMFIO.

In the exam: always write the number table (letter, value, calculation, mod 26, result) — method marks even if one letter is wrong.